Linear algebra – 12

Mixed problems: commutators, traces and the numerical range

Homework

12 сентября 2026 года

Theoretical minimum. The trace form $\operatorname{tr}(XY^{\mathsf{T}})$ turns the space of matrices into a Euclidean space, and a hypothesis quantified over all X is a statement about orthogonality in it. A matrix is a commutator exactly when its trace vanishes, so taking traces is the standard way to destroy commutators; conversely an identity involving $AB-BA$ usually forces nilpotency rather than vanishing. Schur's theorem triangularises unitarily and is the right opening move whenever a spectral hypothesis meets an inequality. The numerical range is convex and contains the spectrum. For real matrices, non-real eigenvalues come in conjugate pairs, which is why determinants of real matrices satisfying algebraic identities are so often forced to be non-negative. And when nothing else works: apply the identity to an eigenvector, or restrict to a 2×2 block, or read the whole hypothesis as a linear map on M_n.

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1. (AKHIMO 2018) Let $n \geqslant 2$ be an integer and denote by $A_1, \ldots, A_n \in M_2(\mathbb{C})$ all solutions of the equation $X^n = \begin{pmatrix} 2 & 3 \\ 4 & 6 \end{pmatrix}$. Prove that $\sum_{k=1}^{n} \operatorname{tr} A_k = 0$.

2. (AKHIMO 2022) Let $A, B, C \in M_n(\mathbb{R})$ be skew-symmetric matrices with $\det(A^2 + B^2 + C^2) = 0$. Prove that $\det(AP + BQ + CR) = 0$ for every triple $P, Q, R \in M_n(\mathbb{R})$.

3. (SEEMOUS 2025) Let A be an $n \times n$ matrix with strictly positive entries and let $u, v \in \mathbb{R}^n$ have strictly positive entries with $Au = v$ and $Av = u$. Prove that $u = v$.

4. (SEEMOUS 2024) Let $A, B \in M_n(\mathbb{R})$ be symmetric with non-negative eigenvalues. Prove that $A^3 + B^3 = (A+B)^3$ if and only if $AB = O_n$.

5. (SEEMOUS 2023) Prove that if A and B are complex $n \times n$ matrices satisfying

$$A = AB - BA + A^2B - 2ABA + BA^2 + A^2BA - ABA^2,$$

then $\det A = 0$.

6. (SEEMOUS 2021) Let $n \geqslant 2$ and let $A \in M_n(\mathbb{R})$ satisfy $A^2 = -E$. If $B \in M_n(\mathbb{R})$ and $AB = BA$, prove that $\det B \geqslant 0$.

7. (SEEMOUS 2017) Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{R})$ satisfy $a^2+b^2+c^2+d^2 < \frac{1}{5}$. Show that $E + A$ is invertible.

8. (SEEMOUS 2016) Let $A_1, \ldots, A_k \in M_n(\mathbb{R})$ be idempotent. Prove that

$$\sum_{i=1}^{k} N(A_i) \geqslant \operatorname{rank}\Big(E - \prod_{i=1}^{k} A_i\Big), \qquad \text{where } N(A_i) = n - \operatorname{rank} A_i.$$

9. (AKHIMO 2022) Let $A = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ with real λ_i. Prove that there exist a unitary matrix U and a real number λ such that $|AU - UA| \geqslant |A - \lambda E|$, where $|B| = \sqrt{B^*B}$ and $X \geqslant Y$ means that $X - Y$ is positive semidefinite.

10. (SEEMOUS 2015) For an integer $n > 2$ let $A, B, C, D \in M_n(\mathbb{R})$ satisfy $AC - BD = E$ and $AD + BC = O$. Prove that

    a) $CA - DB = E$ and $DA + CB = O$;

    b) $\det(AC) \geqslant 0$ and $(-1)^n \det(BD) \geqslant 0$.

11. (IMC) Let $A, B \in M_n(\mathbb{C})$ satisfy $A^2B + BA^2 = 2ABA$. Prove that $(AB - BA)^k = O$ for some positive integer k.

12. (VJIMC 2010) Let A, B be complex 2×2 matrices with $AB-BA = B^2$. Prove that $AB = BA$.

13. (IMC) Let A, B be real $n \times n$ matrices with $\operatorname{rank}(AB - BA + E) = 1$. Prove that $\operatorname{tr}(ABAB) - \operatorname{tr}(A^2B^2) = \frac{1}{2}n(n-1)$.

14. (IMC) Let A, B, C be real square matrices of the same size with A invertible. Prove that if $(A - B)C = BA^{-1}$, then $C(A - B) = A^{-1}B$.

15. Prove the Toeplitz–Hausdorff theorem: the numerical range $W(A)$ is convex. Deduce its real version: for real symmetric A, B and $n \geqslant 3$ the set $\{(\langle Ax, x\rangle, \langle Bx, x\rangle) : \|x\| = 1\}$ is convex.

16. (after IMC 2025 shortlist) Let A, B be real $n \times n$ matrices with $\operatorname{tr}\big((AB)^2\big) = \operatorname{tr}(A^2B^2)$.

    a) Prove that if all eigenvalues of $AB - BA$ are real, then $AB - BA$ is nilpotent.

    b) Show that for $n = 2$ the assumption on the eigenvalues can be dropped, but for every $n \geqslant 3$ it cannot.